

Topological Quantum Matters (TQM)

Time:

Every Mon. & Wed. 13:30-15:05, from 2021-9-13 to 12-3 (weeks 1-12 of the fall semester in Tsinghua)

Venue:

It will be a combination of offline (宁斋W11, Ning Zhai W11) and online (腾讯会议tencent meeting: 5772849861, password: 654321)

Description:

In this course, we will use topology to understand some exotic quantum phases of matter. The topics will include topological insulators, topological orders, symmetry-protected topological phases, etc. The course will cover both condensed matter models in physics and general mathematical descriptions (such as group cohomology theory and modular tensor categories of knots).

Prerequisites:

Basic topology and quantum mechanics. We will try to make a compromise between mathematics and physics by introducing relevant concepts self-consistently, as there are audience from both sides.

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Other courses:

YMSC { Reshetikhin : Inv. of knots and 3-mfd.
Hashigahara : Topological Quantum Computation.
Zheng : Category theory

BIMSA { Papadimitriou : Quantum field theory anomalies and applications
Palcoux : On fusion categories
Yibing Wang : Modular categories and Reshetikhin-Turaev TQFTs

Introduction to QM.

c) Why important to real life?

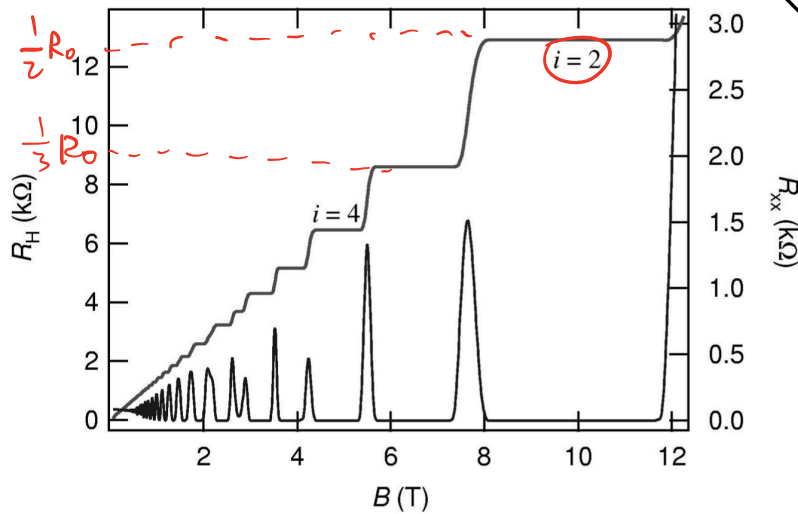
One example:

Q: how to measure { electric charge e
Planck constant h accurately? }

① Integer
Quantum Hall effect

Nobel 1985

$$R_0 = \frac{h}{e^2}$$



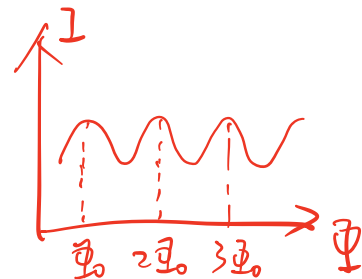
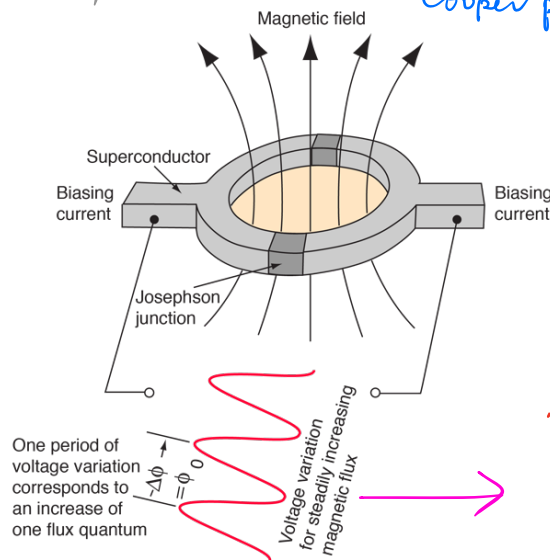
$$\begin{cases} e = \frac{2 \Phi_0}{R_0} \\ h = \frac{4 \Phi_0^2}{R_0} \end{cases}$$

② Josephson effect

Nobel 1973

$$\Phi_0 = \frac{h}{2e}$$

Cooper pair



International
System of
Units
(before 2018)

[After 2018, e, h are fixed by definition]

using many-body system to measure properties of a single electron!
accurately

(2) Why exotic?

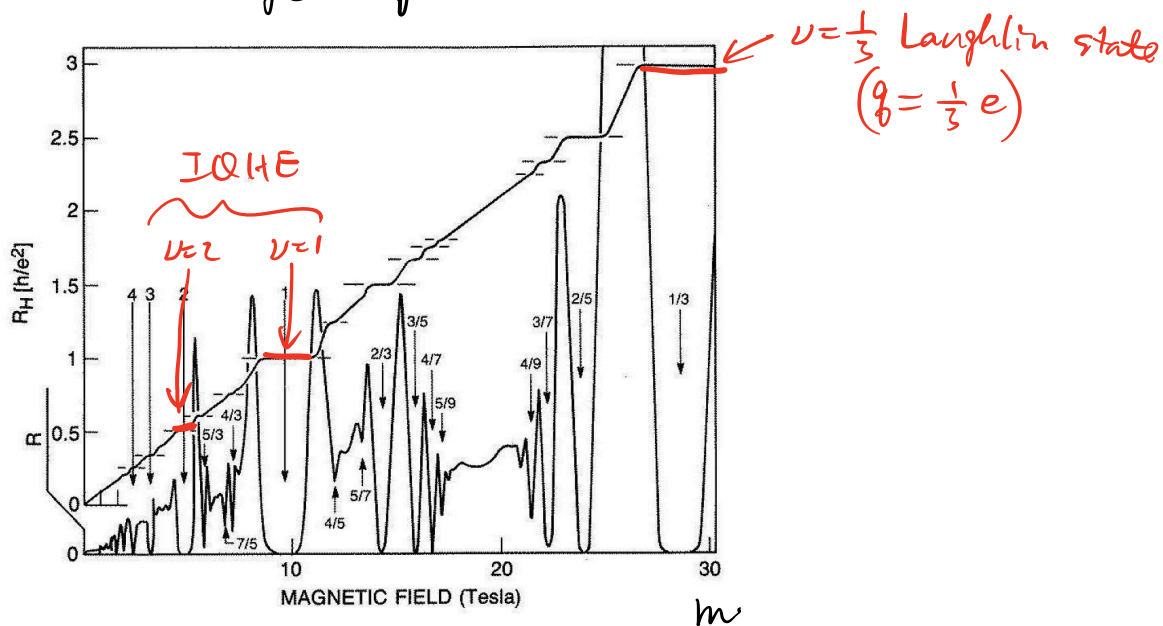
fundamental charge e = charge of one single electron/proton/...

Fractional quantum Hall effect (FQHE) Nobel 1998

$$R = \frac{h}{\nu e^2}$$

$$\begin{cases} v \in \mathbb{Z} : & \text{integer QHE} \\ v \in \mathbb{Q} : & \text{FQHE} \end{cases}$$

Fractional charge $q = v \cdot e$



Fractional charge appears in many-body systems!

(3) Why interesting to many people ?

related to many other topics:

math: knot theory, Jones polynomial, Turaev-Viro inv.,
Reshetikhin-Turaev inv., modular tensor categories, ^{Zheng's lecture}
2-knot, higher categories ...

physics: IQHE, FQHE, topological insulator, ...
(real materials)

quantum information: topological quantum computation
(by Freedman, Kitaev)
fault-tolerant quantum computation

(4) Why hard/deep?

$\propto$ number of degrees of freedom

$\propto$ dimensional space.

foundations of quantum field theory (= many-body quantum mechanics)

emergence of spacetime/gravity (?)

Syllabus (tentative):

It may vary depending on the actual speed of the course. By weeks (4*45min/week, 12 weeks):

- (1) introduction to TQM, different classes of TQM (bosonic/fermionic, long/short range entangled, with/without symmetry),
1st example: Kitaev's toric code model (homology enters),
2nd example: Haldane's honeycomb model (homotopy enters) *1988* $\rightsquigarrow$ *Qikun Xue : quantum anomalous Hall effect. 2013*

Part I. Bosonic topological orders

- (2) quantum double model, twisted quantum double model = Dijkgraaf-Witten gauge theory
- (3) introduction to fusion categories, Levin-Wen model = Turaev-Viro model
- (4) introduction to modular tensor categories, general description of anyon models by Kitaev
- (5) 3+1D Walker-Wang model = Crane-Yetter model

Part II. Topological insulators (fermionic symmetry-protected topological phases without interactions)

- (6) introduction to band theory, integer quantum Hall effect, Thouless-Kohmoto-Nightingale-den Nijs number, Chern insulator
TKNN : Hall conductance = Chern number
- (7) examples: Kitaev's Majorana chain, Su-Schrieffer-Heeger model, 2+1D and 3+1D topological insulators, edge theories
- (8) symmetries in free fermion system, 10-fold way classification $\rightarrow$ *topological K theory, Clifford algebras*

Part III. Symmetry-protected topological phases

- (9) introduction to symmetry-protected topological (SPT) phases, Haldane chain
- (10) introduction to projective representation, tensor product state, classification of 1+1D bosonic SPT
- (11) Levin-Gu model, introduction to group cohomology, bosonic SPT model from group cohomology
- (12) introduction to fermionic SPT phases, other related topics

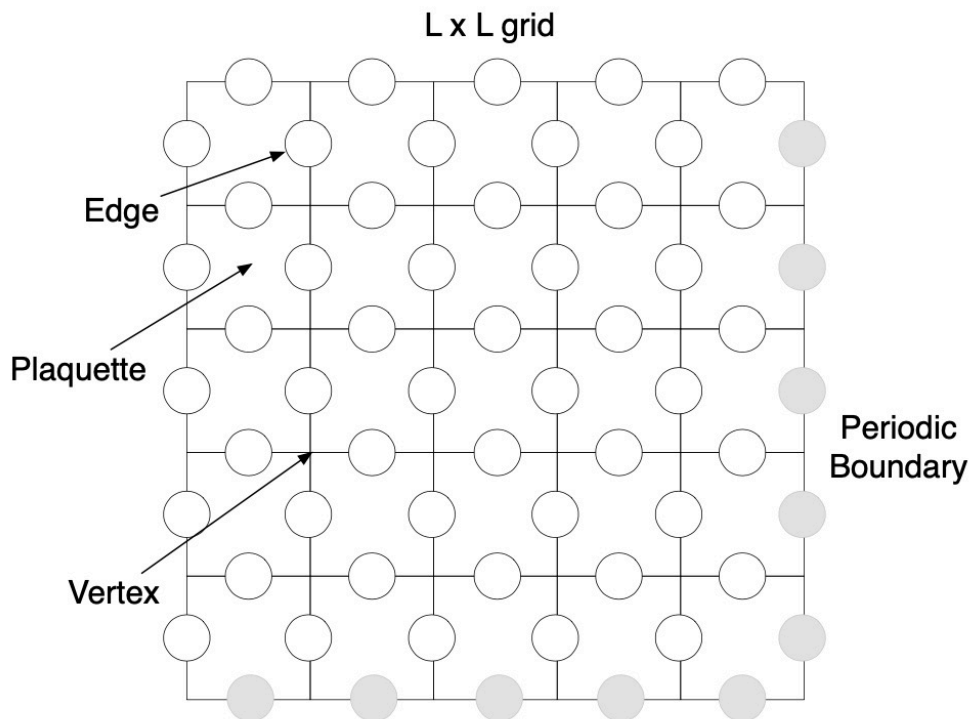
1st example of TQM:

Toric code model $\left(= \mathbb{Z}_2 \text{ gauge theory} \right)$
(by Kitaev 1997) on torus

Hilbert space on torus

$L \times L$ square lattice, $\mathbb{C}^2$ (spin 1/2) on each link

$$\mathcal{H} = \bigotimes_{\text{link } j} \mathbb{C}^2$$



$\mathcal{H}$: Hilbert space

Hamiltonian $H: \mathcal{H} \rightarrow \mathcal{H}$

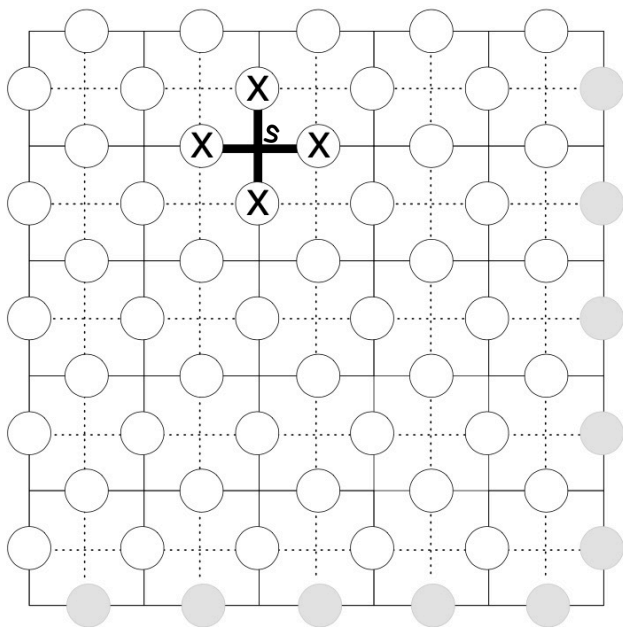
eigenvalues of H are energies E_n of the system

Ground state is the eigenvector with the lowest energy E_0 .

Claim $\left\{ \begin{array}{l} \bullet \text{ The ground state of TC represents } H_1(T^2, \mathbb{Z}_2) = \mathbb{Z}_2 \times \mathbb{Z}_2 \\ \bullet \text{ robust ground states} \\ \bullet \text{ Topological quantum information} \end{array} \right.$

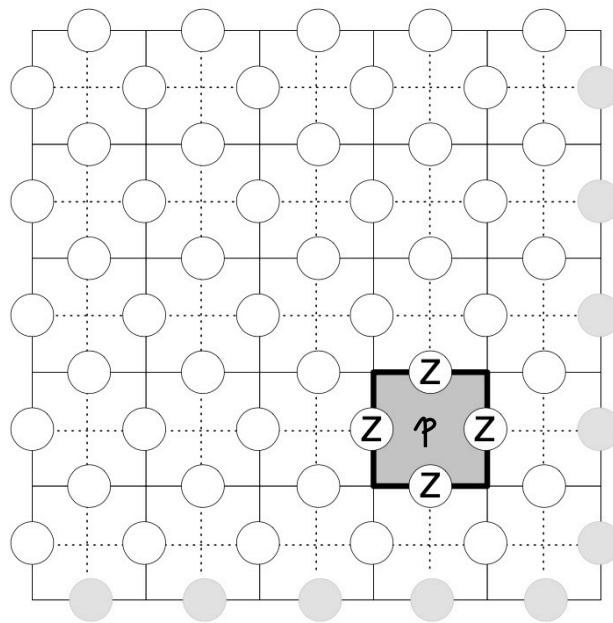
Hamiltonian

$$H = - \sum_{\text{site } s} A_s - \sum_{\text{plaquette } p} B_p$$



$$A_s = \prod_{j \in \text{star}(s)} \sigma_j^x$$

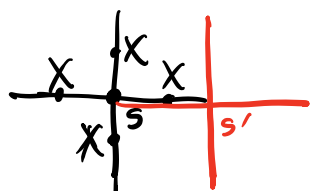
Pauli operator
 $X = \sigma^x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$



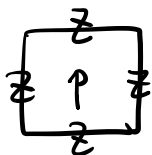
$$B_p = \prod_{j \in \partial p} \sigma_j^z$$

$Z = \sigma^z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

Algebraic relations of A_s, B_p .

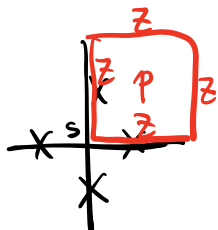


$$\begin{cases} A_s^\dagger = A_s \\ (A_s)^2 = 1 \\ [A_s, A_{s'}] = A_s A_{s'} - A_{s'} A_s = 0 \end{cases} \quad \leadsto A_s \text{ has eigenvalues } \pm 1$$



$$\begin{cases} B_p^\dagger = B_p \\ (B_p)^2 = 1 \\ [B_p, B_{p'}] = 0 \end{cases} \quad \leadsto B_p \text{ has eigenvalues } \pm 1$$

$$[A_s, B_p] = A_s B_p - B_p A_s = 0.$$



$$XZ = -ZX \Rightarrow A_s B_p = B_p A_s$$

$$H = - \sum_s A_s - \sum_p B_p$$

Ground state $|\Psi\rangle$ of TC:

$$A_s |\Psi\rangle = B_p |\Psi\rangle = +1 |\Psi\rangle, \quad \forall s, p.$$

String representation.

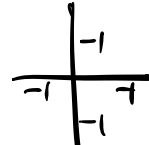
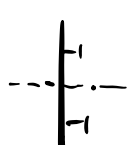
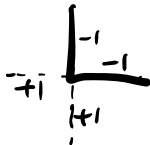
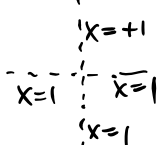
$$A_s = \prod \sigma^x.$$

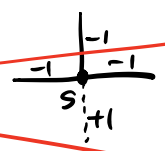
$$\sigma^x: \pm 1$$

$$\text{def: } \begin{cases} \sigma_j^x = -1 : \text{ string} \\ \sigma_j^x = +1 : \text{ no string} \end{cases}$$

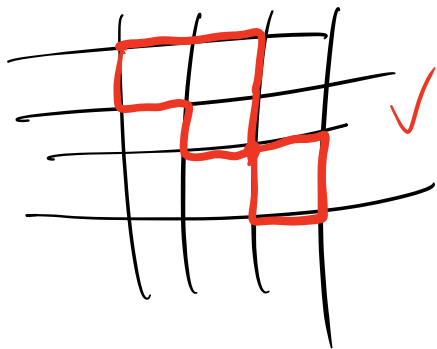


low energy: $A_s = 1$:

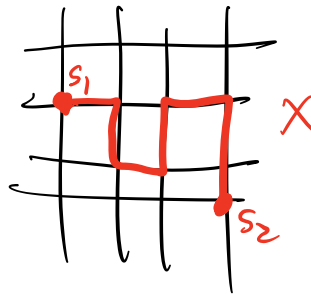


high energy: $A_s = -1$: 

ground state: $A_s |\Psi\rangle = |\Psi\rangle \iff$ closed strings (loops)

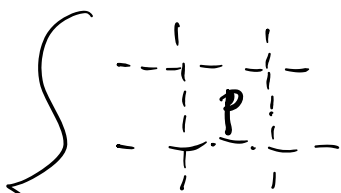


$A_s = +1$
($\forall s$)

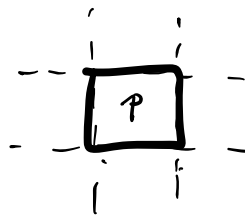


$A_{s_1} = A_{s_2} = -1$
(high energy)

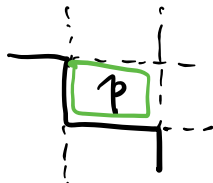
$B_p = \pi \mathbb{Z}$: flip $\sigma^x = +1 \leftrightarrow \sigma^x = -1$



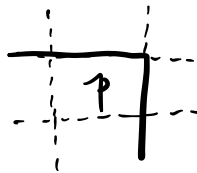
$\xleftrightarrow{B_p}$



create
and/or
small loops.



$\xleftrightarrow{B_p}$



change shapes
of closed loops.

$B_p |c\rangle = |c + \partial p\rangle \pmod{2}$
 $\downarrow$
 $\forall$ closed loop configuration

$$H = - \sum A_s - \sum B_p$$

$\downarrow$ closed loop. $\downarrow$ change shapes.

Assume $|\Psi\rangle = \sum_c a_c |c\rangle$, $a_c \in \mathbb{C}$ is the ground state

$$\begin{aligned} \Rightarrow B_p |\Psi\rangle &= \sum_c a_c |c + \partial p\rangle = \sum_c a_{c + \partial p} |c\rangle \\ &= |\Psi\rangle = \sum_c a_c |c\rangle \end{aligned}$$

$$\Rightarrow a_c = a_{c + \partial p} \text{ for } \forall p$$

The ground state coefficients of $|c\rangle$ and $|c + \partial p\rangle$ should be the same.
 or Loops that are homologous to each other have the same coefficient.

Summary { $A_S = +1$: conf. $|c\rangle$ such that there are only closed loops in c .
 $B_P = +1$: $|c\rangle$ and $|c + \partial p\rangle$ have the same coefficient.
 (mod 2)
 enforce closed-loop constraints
 mod out homologous loops.

Ground state subspace of TC

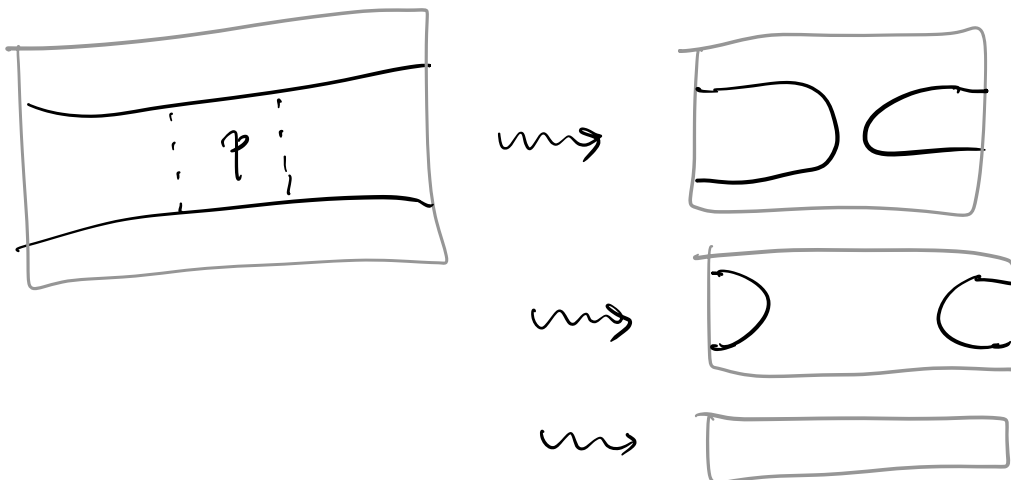
= { closed loops } / { boundaries of 2D surfaces on torus }

$$= H_1(T^2, \mathbb{Z}_2) = \mathbb{Z}_2 \times \mathbb{Z}_2$$

$$|\psi\rangle = | \quad \rangle + | 0_0^0 \rangle + | \bigcirc \rangle + | \text{---} \rangle + \dots$$

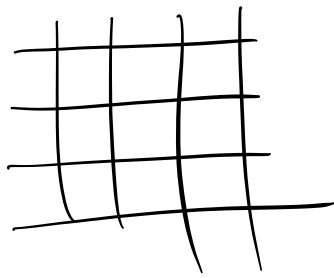
4-fold degeneracy of TC on torus :

$$\begin{cases} |\psi_{00}\rangle = | \quad \rangle + | 0_0^0 \rangle + | 00 \rangle + \dots \\ |\psi_{10}\rangle = | \text{---} \rangle + | \frac{0}{0} \rangle + | \bigcirc \rangle + \dots \\ |\psi_{01}\rangle = | | \rangle + | \frac{0}{0} \rangle + | 0|0 \rangle + \dots \\ |\psi_{11}\rangle = | \diagup \rangle + | \diagdown \rangle + | \frac{0}{0} \rangle + \dots \end{cases}$$



Last time:

TC:



towns

square lattice

$$|1\rangle = - \sum_{\text{site } s} A_s - \sum_{\substack{\text{plaquette} \\ p}} B_p$$

$A_5 =$ 

$$B_1 = \begin{array}{|c|} \hline z \\ \hline p \\ \hline z \\ \hline \end{array}$$

$$[A_s, B_p] = 0$$

$$(A_S)^2 = 1$$

$$\left(\frac{1+A_S}{2}\right)^2 = \underbrace{\frac{1+A_S}{2}}_{\text{projector to } A_S=1 \text{ states}}$$

commuting - projector Hamiltonian.

$A_s = +1$ for every site s of the torus.
 $\hookrightarrow$ closed string property of the GS.
 $(\sigma_j^x = -1 \text{ for link } j)$

$$B_p = \{1\} :$$

$$B_p \left| \text{loop with } p \right\rangle = \left| \text{loop with } p \right\rangle = \left| \text{loop with } p \right\rangle$$

GS of TC: $A_S |\psi\rangle = B_P |\psi\rangle = |\psi\rangle$, $\forall s, p$.

$$|\psi\rangle = \sum_c a_c |c\rangle, \quad a_c \in \mathbb{C}$$

$$\begin{cases} A_s |\psi\rangle = |\psi\rangle \Rightarrow c \text{ is closed loop conf.} \\ B_p |\psi\rangle = |\psi\rangle \Rightarrow B_p \sum_c a_c |c\rangle = \sum_c a_c \underbrace{|c + \vec{ap}\rangle}_{(\text{mod } 2)} \\ = \sum_c a_{c+\vec{ap}} |c\rangle \\ = \sum_c a_c |c\rangle \end{cases}$$

$$\Rightarrow a_{c+ap} = a_c, \forall p$$

$$\Rightarrow a: \mathcal{H}|_{A_s=1} \rightarrow \mathbb{C}$$

$$c \mapsto a_c$$

is a function of $\mathbb{Z}_2$ homology class of c .

Different classes of TQM:

① bosonic / fermionic

{ Microscopic degrees of freedom.
Not emergent excitations.

Example: bosonic Toric code has emergent fermions.

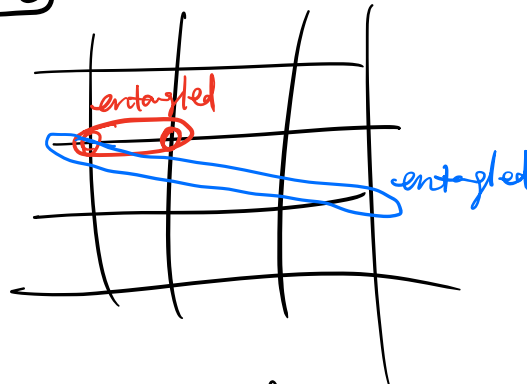
② long range entangled / short range entangled
(LRE) (SRE)

$$\text{entanglement } |\Psi\rangle = |\uparrow\uparrow\rangle + |\downarrow\downarrow\rangle$$

(in QM)

$\downarrow \quad \downarrow$
 1st 2nd particle

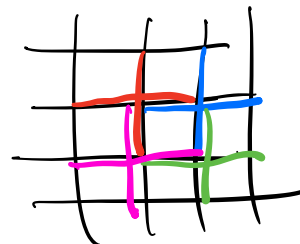
Add distance structure:



Example: T.C. is long range entangled.

$$A_s = \begin{array}{c} \times \\ \times \times \times \\ \times \end{array}$$

$$\prod_s A_s = 1 \text{ on torus}$$



$$A_s = -1 \rightarrow A_i = -1 \text{ for } s' \in \bar{i}$$



measure LRE: topological entanglement entropy.

$$S = \underbrace{(\text{area law})}_L + \underbrace{S_{\text{top.}}}_{L^0} + \underbrace{\frac{\text{const.}}{L}}_{L^{-1}} + \dots$$

③ with / without symmetries

Ising $\mathbb{Z}_2$ symm.

$SU(2)$, $U(1)$

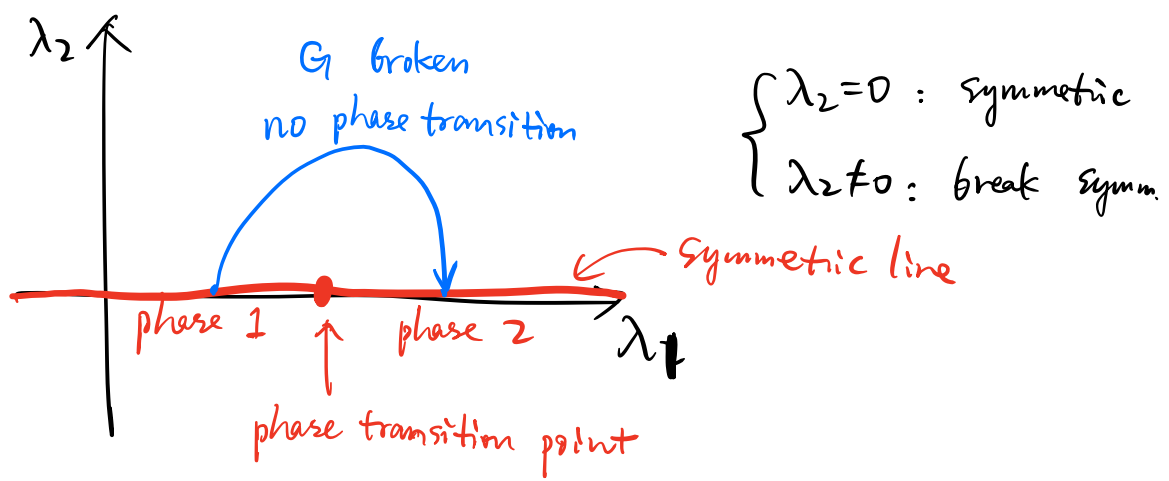
time reversal.

...

Given a symmetry group G , there may be many different topological phases. If we can break G , they may become the same phase.

G	3	4
phase 1	...	
2		

phase diagram



① bosonic / fermion

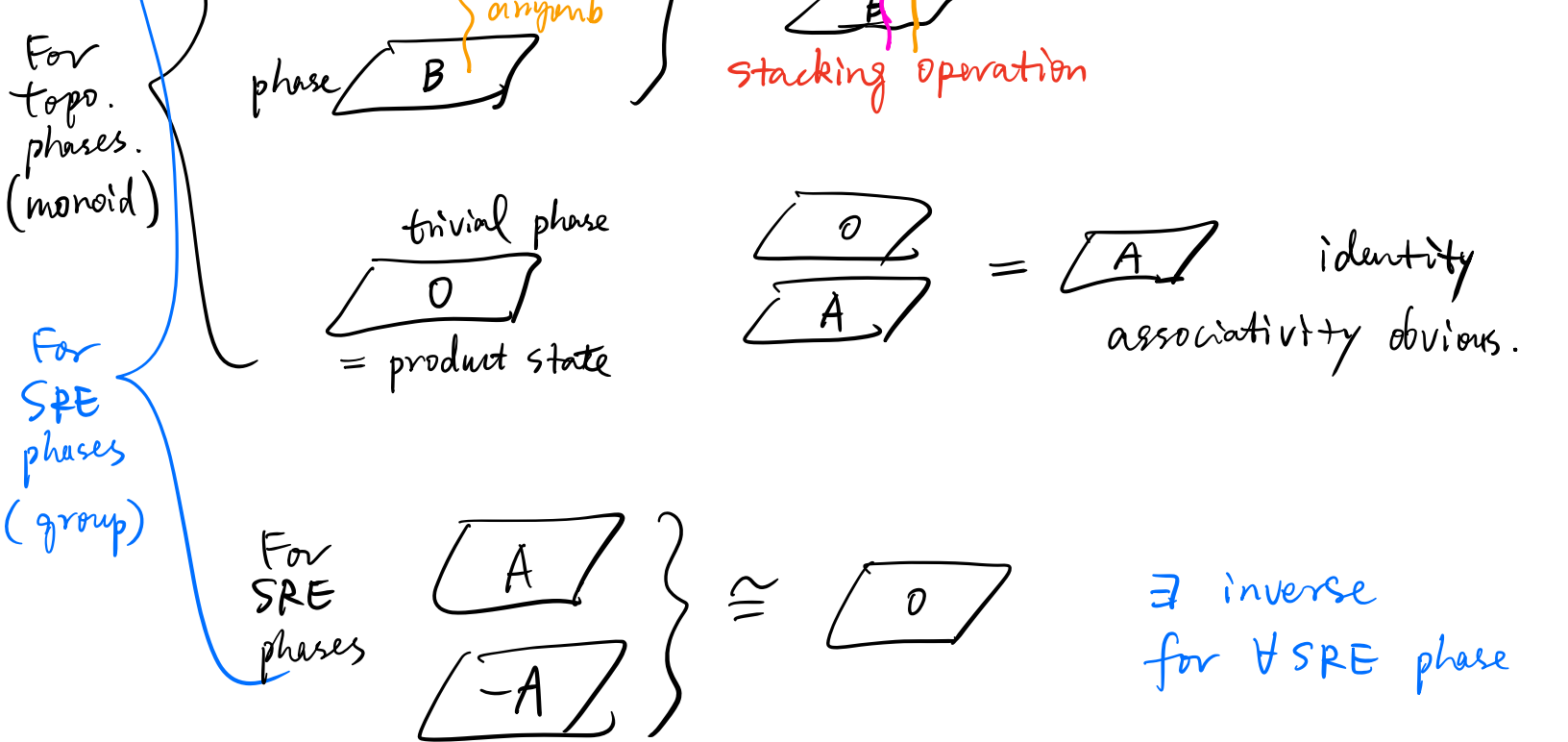
③ \ ②	②	
	SRE	LRE
No symm.	invertible topo. order (iTO)	topo. order (TO)
Symm.	Symmetry-protected topo. order (SPT)	Symmetry-enriched topo. order (SET)

Statement :

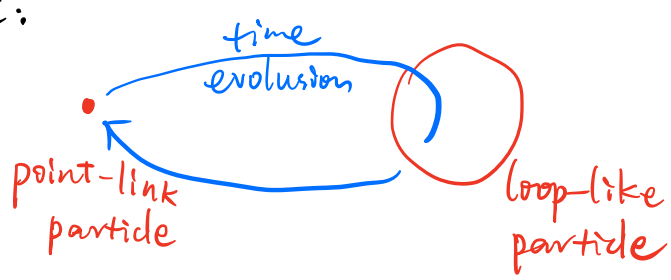
① The collection of all topological phases forms a Abelian monoid (=group without inverse)

② The collection of all SRE phases has an Abelian group structure.

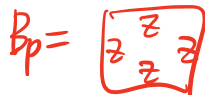
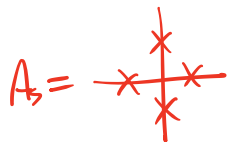




3+1D TC:



$$H = - \sum_s A_s - \sum_p B_p$$



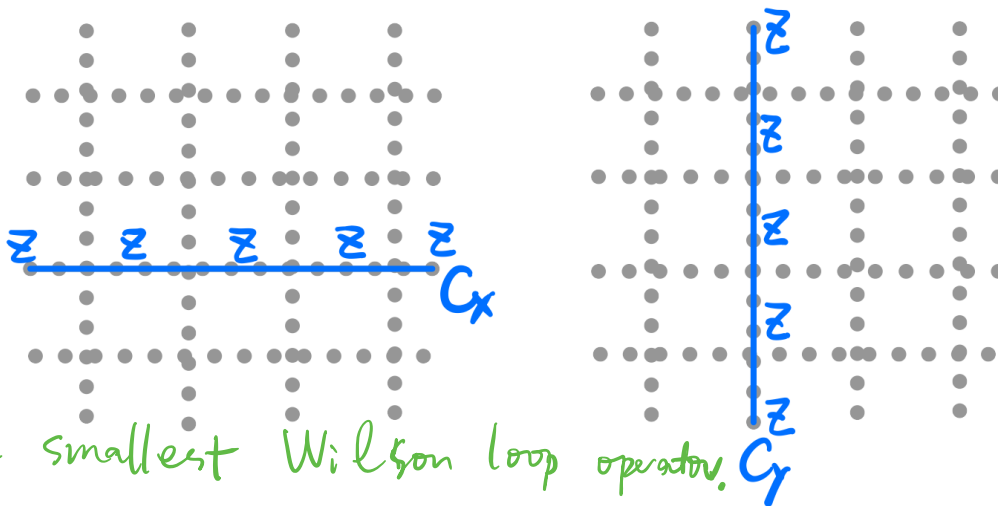
Wilson operators

U(1) gauge theory
 $W = e^{i \int_C A_\mu dx^\mu}$

For a loop C , define Wilson loop operator

$$W_C = \prod_{j \in C} \sigma_j^z, \text{ the ground states are related by}$$

$$W_{C_x}, W_{C_y}$$



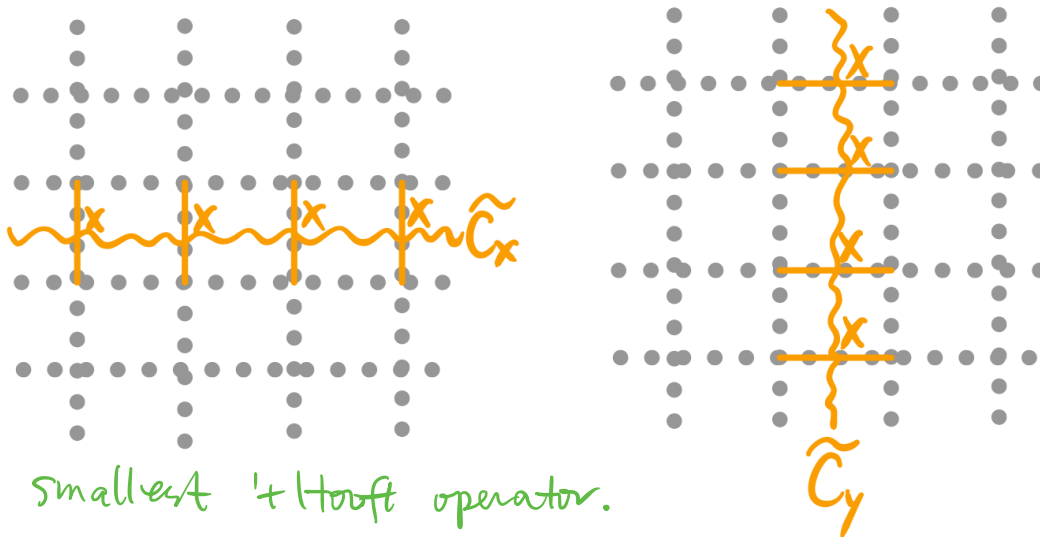
B_p is the smallest Wilson loop operator.

't Hooft operators

For a dual loop C , define 't Hooft operator

$V_{\tilde{C}} = \prod_{j \perp \tilde{C}} \sigma_j^x$, the ground states are related by

$$V_{\tilde{C}_x}, V_{\tilde{C}_y}$$



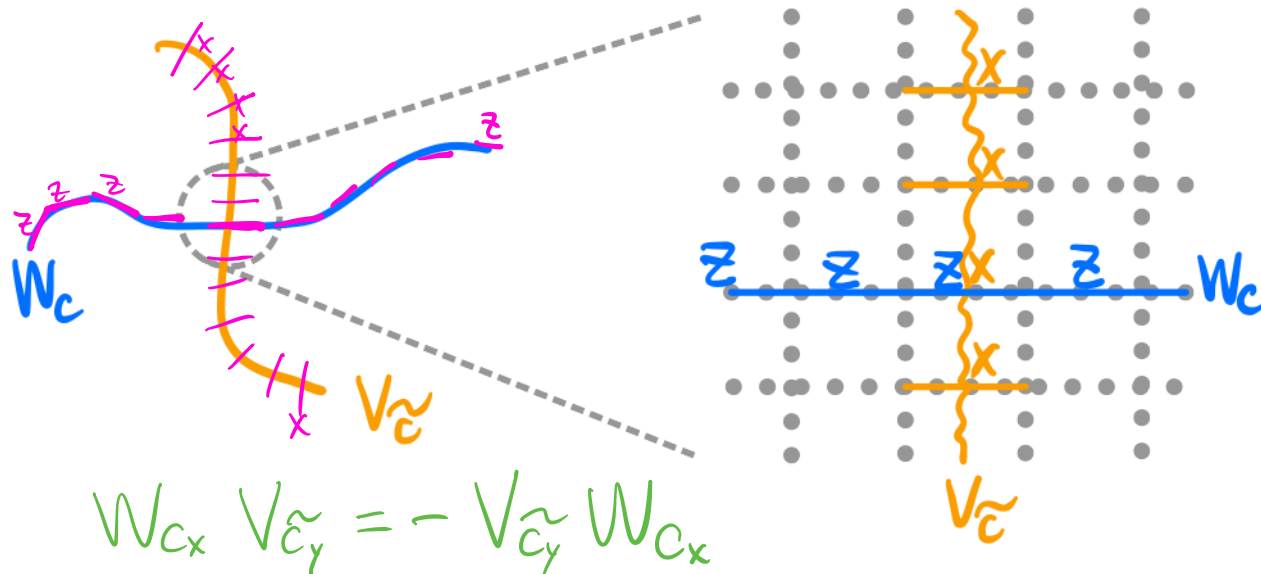
As is the smallest 't Hooft operator.

Heisenberg algebra from W and V

For a loop C and dual loop $\tilde{C}$, we have

$$W_C V_{\tilde{C}} = (-1)^{\#C \cap \tilde{C}} V_{\tilde{C}} W_C$$

which has smallest representation $\dim 4$.



e excitations

Endpoints of Wilson operator W_C are electric charge excitations e .

$$A_s |GS\rangle = B_p |GS\rangle = |GS\rangle$$

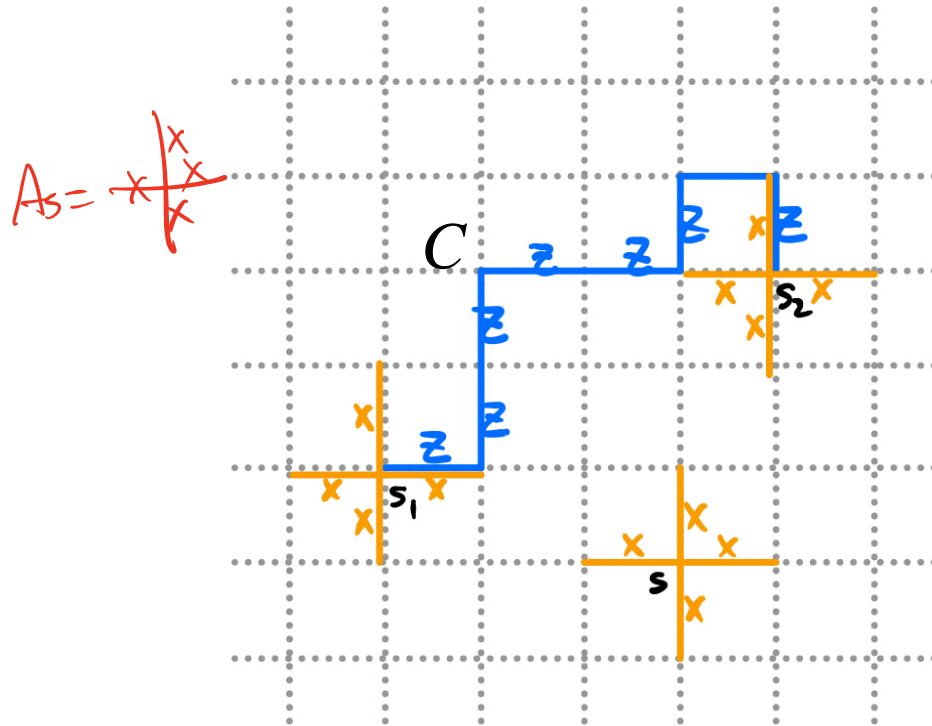
$$|e_{s_1} e_{s_2}\rangle = W_C |GS\rangle$$

$$A_{s_2} W_C |GS\rangle = -W_C A_{s_2} |GS\rangle = -W_C |GS\rangle$$

$$\Rightarrow A_{s_1} = A_{s_2} = -1$$

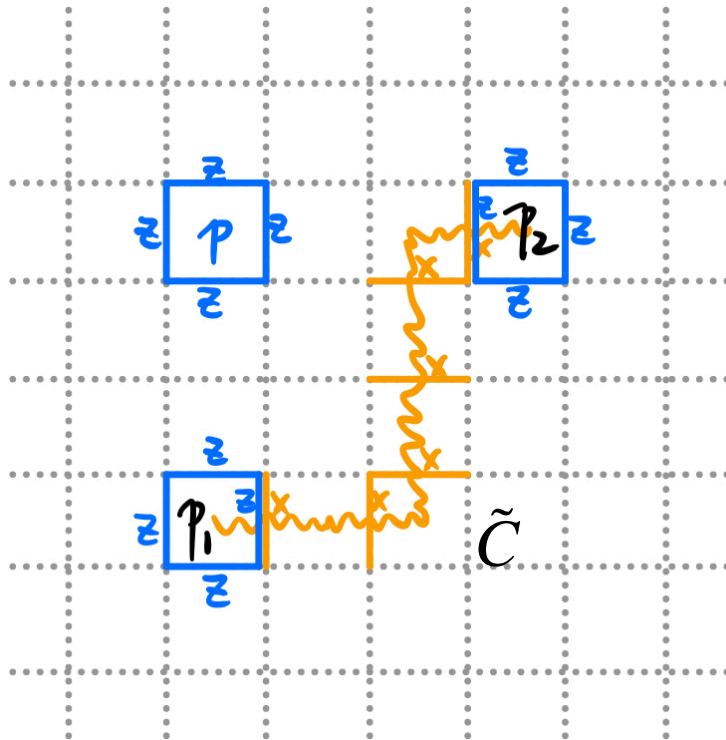
$$A_{s \neq s_1, s_2} = 1$$

$$B_p = 1$$



m excitations

Endpoints of 't Hooft operator $V_{\tilde{C}}$ are magnetic charge excitations m .



$$|m_{p_1} m_{p_2}\rangle = V_{\tilde{C}} |\text{GS}\rangle$$

$$B_{p_1} = B_{p_2} = -1$$

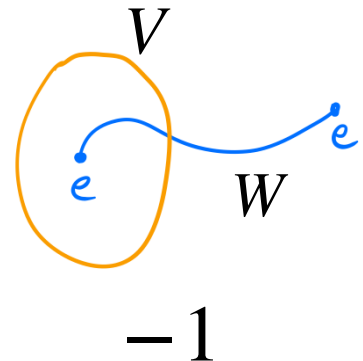
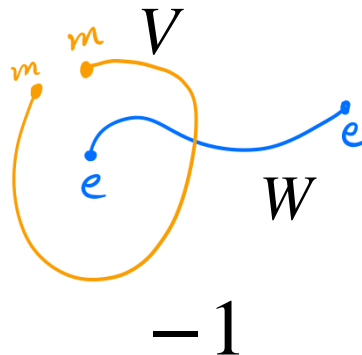
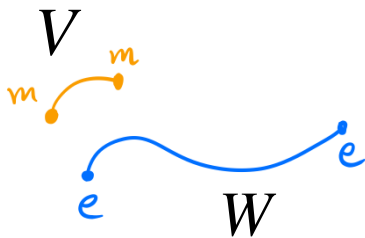
$$B_{p \neq p_1, p_2} = 1$$

$$A_s = 1$$



Statistics

- W_C and $V_{\tilde{C}}$ can be used to create / move e and m excitations
- Full braiding of e and m gives -1 sign



Anyons

- In (2+1) dimensions, point-like excitations (**anyons**) can have any rational number statistical phases.

$$f = \left(\begin{smallmatrix} e \\ m \end{smallmatrix} \right) \xrightarrow{\frac{W_C}{V_C}} \text{ribbon operator (with framing)}$$

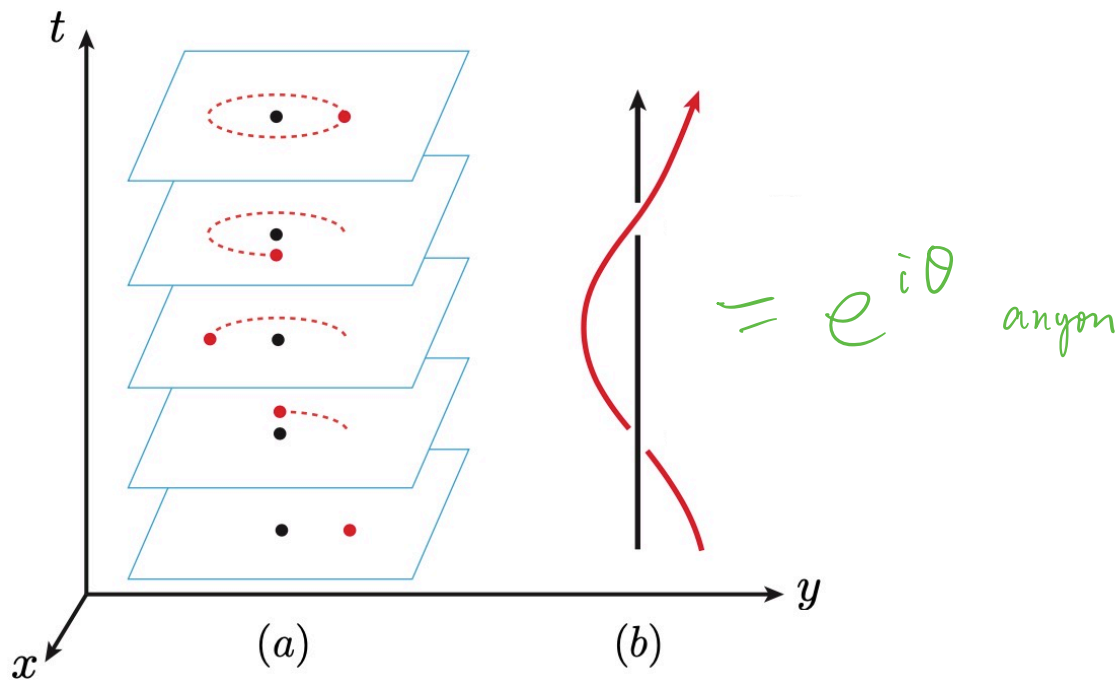
- Anyons in toric code: **$1, e, m, f = em$** (with framing)
- Both e and m are bosons; f is fermion
- e, m, f have mutual braiding phase -1

boson/fermion: $e_1 \cdot e_2 \Rightarrow \pm e_2 \cdot e_1$

$$\begin{array}{c} \text{X} \\ \text{e}_1 \text{ m}_1 \quad \text{e}_2 \text{ m}_2 \end{array} = - \begin{array}{c} \text{parallel lines} \\ \text{e}_1 \quad \text{e}_2 \end{array}$$

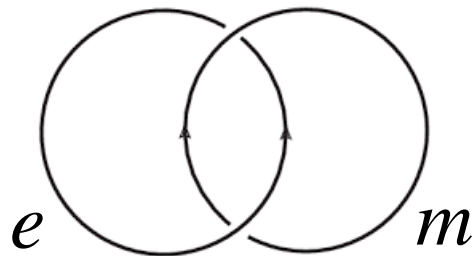
knot/link in (2+1)D spacetime

anyon worldlines in (2+1)D $\longleftrightarrow$ knot/link in 3D



Hopf link

For the toric code model, mutual statistics of e and m :


$$= -1$$

In general, for a link of e and m worldlines, we have statistical phase $(-1)^{\text{linking \#}}$.

Anyon models $\rightarrow$ knot/link invariants

Summary

- Toric code model is a commuting projector Hamiltonian on torus
- Ground state subspace $\cong H_1(T^2, \mathbb{Z}_2)$
- Anyons braiding statistics and knot/link invariants
- Generalizations of toric code model later...